AXIAL PISTON PUMPS / PRODUCT INFORMATION

P60 Axial Piston Pump

Check the larger flow duty together with supply-line losses and inlet configuration. Use the published curves at matching speed and pressure; separate maximum flow and pressure values do not establish one simultaneous operating point.

The P60 more than doubles the flow of the P30, reaching 146 L/min (38.6 US gpm) at 70cc (4.27 in³) and 1750 rpm, while simultaneously achieving peak overall efficiencies above 89% at 1500 rpm. At this displacement and body size, internal leakage as a proportion of total flow is small enough that the P60 maintains efficiency above 87% across a wide band from 60 to 140 bar (870 to 2,031 psi) at 1500 rpm, making it a genuinely high-efficiency machine for sustained industrial duty.



WaterQuest P60 axial piston water hydraulic pump, 316 stainless steel, 200mm body

At a glance

Parameter / value	Parameter / value
Displacements 51/63/70 cc/rev (4.272 in³/rev)	Max Pressure 160 bar (2,321 psi) continuous
Max Flow 146 L/min (38.57 US gpm)	Max Input Power 42 kW (56.32 hp)
Peak Efficiency 89% at 80-120 bar (1,160-1,740 psi)	Body OD 200 mm ()

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	51 cc/rev (3.11 in³/rev)	63 cc/rev (3.84 in³/rev)	70 cc/rev (4.27 in³/rev)	Note
Displacement	51 cc/rev (3.11 in³/rev)	63 cc/rev (3.84 in³/rev)	70 cc/rev (4.27 in³/rev)	Fixed
Max Continuous Pressure	160 bar (2,321 psi)			All configs
Max Flow (unboosted, 1800 rpm)	~88 L/min (23.2 US gpm)	~108 L/min (28.5 US gpm)	~122 L/min (32.2 US gpm)	PC / PA / PPC
Max Flow (boosted, 2000 rpm)	~105 L/min (27.7 US gpm)	~130 L/min (34.3 US gpm)	146 L/min (38.6 US gpm)	PB
Max RPM (Unboosted)	1800 rpm			PC / PA / PPC
Max RPM (Boosted)	2000 rpm			PB
Max Input Power	~28 kW (37.5 hp)	~38 kW (51 hp)	42 kW (56.3 hp)	At 1750 rpm, 160 bar (2,321 psi)
Max Torque at 160 bar (2,321 psi) (1500 rpm)	~143 Nm (105 lbf·ft)	~176 Nm (130 lbf·ft)	~188 Nm (139 lbf·ft)	All configs
Peak Overall Efficiency	~89% at 80-120 bar (1,160-1,740 psi), 1500 rpm (70cc (4.27 in³))			
Max Fluid Temperature	50°C (consult WQS above 50°C (122 °F))			
Min Fluid Temperature	2°C (35.6 °F) (antifreeze option available)			
Housing Material	316 Stainless Steel			
P Port - Unboosted (PC/PA/PPC)	1" BSPP			All displacements
T Port - Unboosted (PC/PA/PPC)	1 1/2" BSPP			All displacements
P Port - Boosted (PB)	1" BSPP (pressure outlet)			All displacements
T Port - Boosted (PB)	1/2" BSPP (case drain to sump)			Max 2.5 bar (36.3 psi) at T port
Shaft Diameter	27.99 / 27.97 mm (1.1 / 1.1 in)			
Shaft Key	8 x 7 mm (0.315 x 0.276 in)			
Flange / Spigot OD	127.00 / 126.95 mm (5 / 5 in)			
Body Outer Diameter	200 mm (7.87 in)			
Mounting Face Diameter	181 mm (7.13 in)			
Overall Body Length	199 mm (7.83 in)			
Shaft Section Length	62 mm (2.44 in) total (45 mm (1.77 in) inner zone)			
Key Projection	12.5 mm (0.492 in)			

Parameter	51 cc/rev (3.11 in³/rev)	63 cc/rev (3.84 in³/rev)	70 cc/rev (4.27 in³/rev)	Note
Base Projection	21.5 mm (0.846 in)			
Port Face	139 mm (5.47 in) square, 36 mm (1.42 in) and 41 mm (1.61 in) spacing			
PPC Mounting Holes	M10 x 15 deep on 155 mm (6.1 in) PCD at 20° spacing			PPC option only
Shaft Rotation	PC = CW, PA = CCW / PPC = CW / PB = bi-directional			
Inlet Filtration Required	10 µm (394 µin) nominal, 25 µm (984 µin) absolute, beta-10 = 75			
Certifications	Confirm applicability and supplied documentation: ATEX II 2GD ck T6, ISO 9001 FM 87247, F4N			

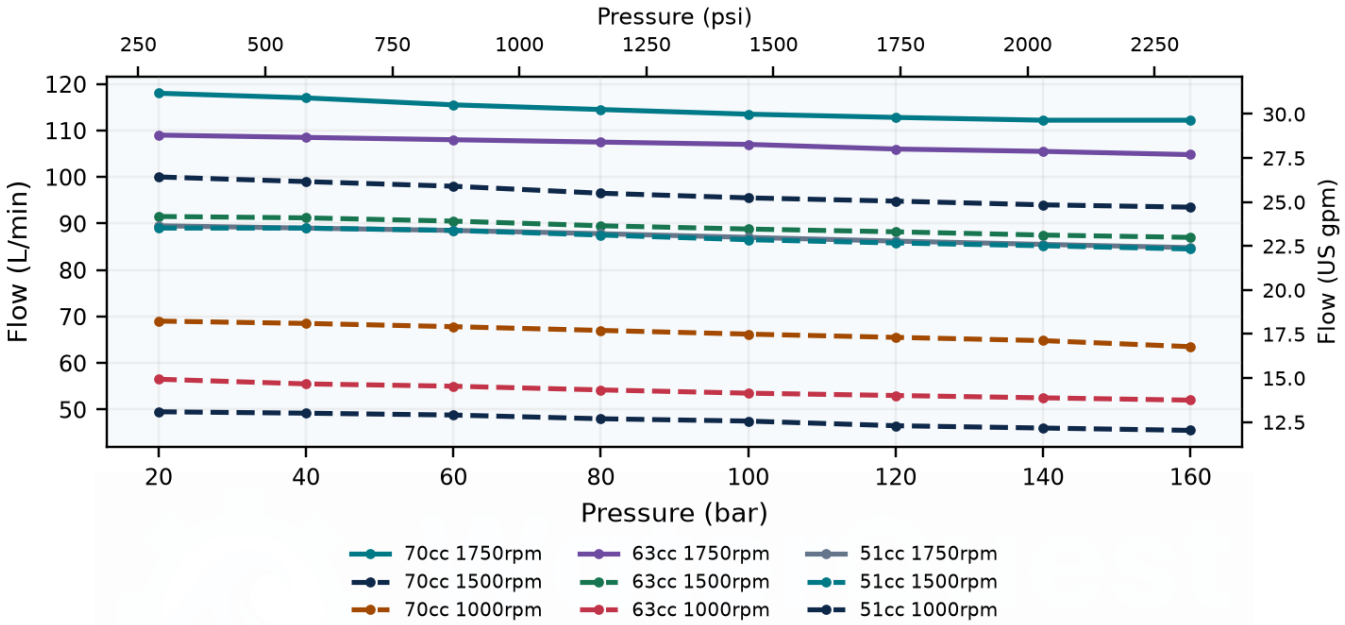
Selection and integration

All performance figures on clean tap water at 15°C (59 °F). The exact operating fluid must be specified on all enquiries and orders. 316 SS Material Test Reports available on request. Contact WaterQuest for motor frame sizing, bell housing selection, and coupling specification at P60 power levels.

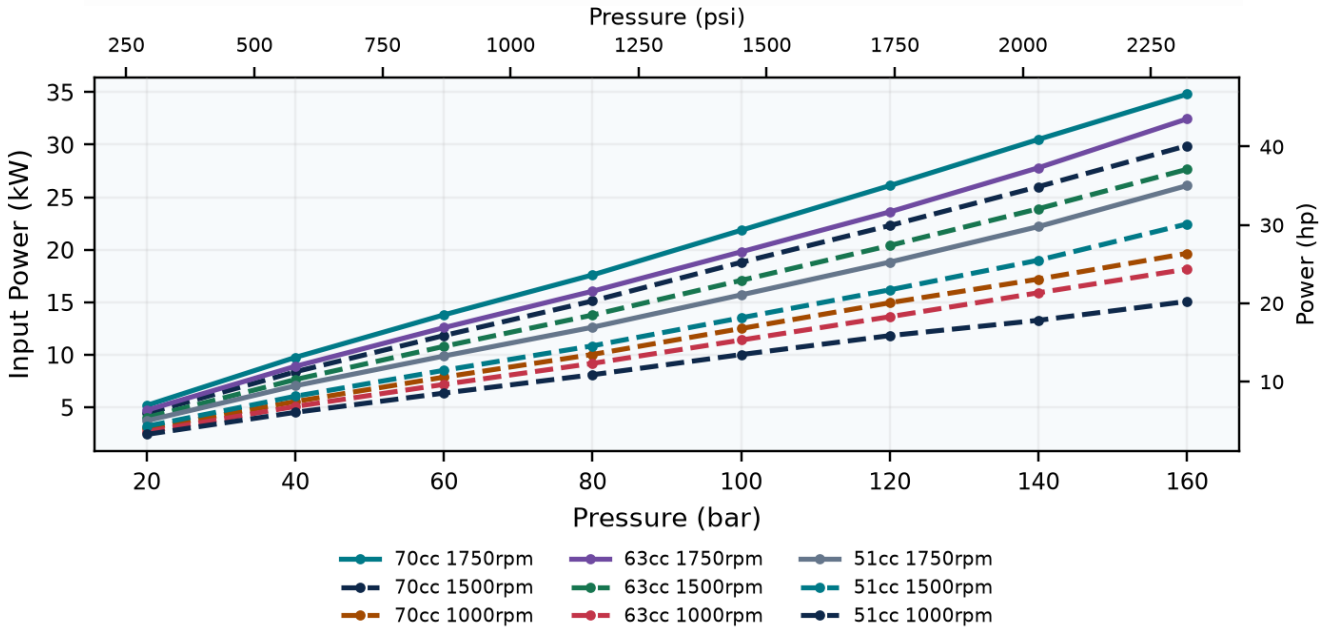
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

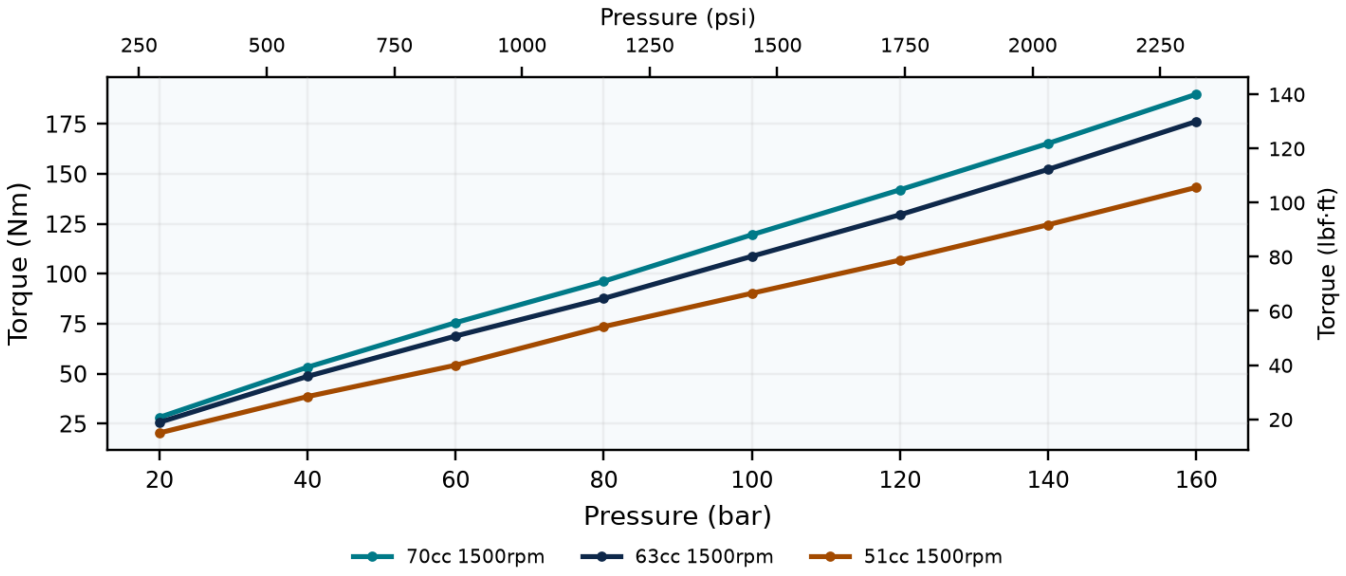
Flow (L/min) vs Pressure (bar)



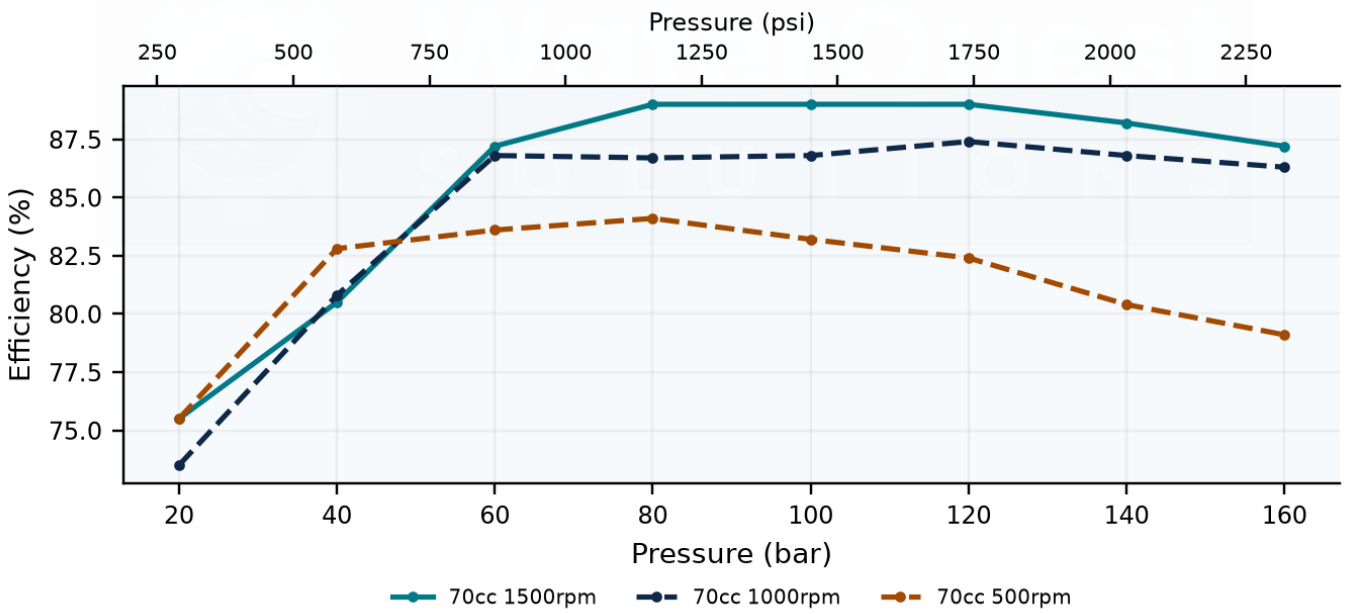
Input Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)



Efficiency (%) vs Pressure (bar)

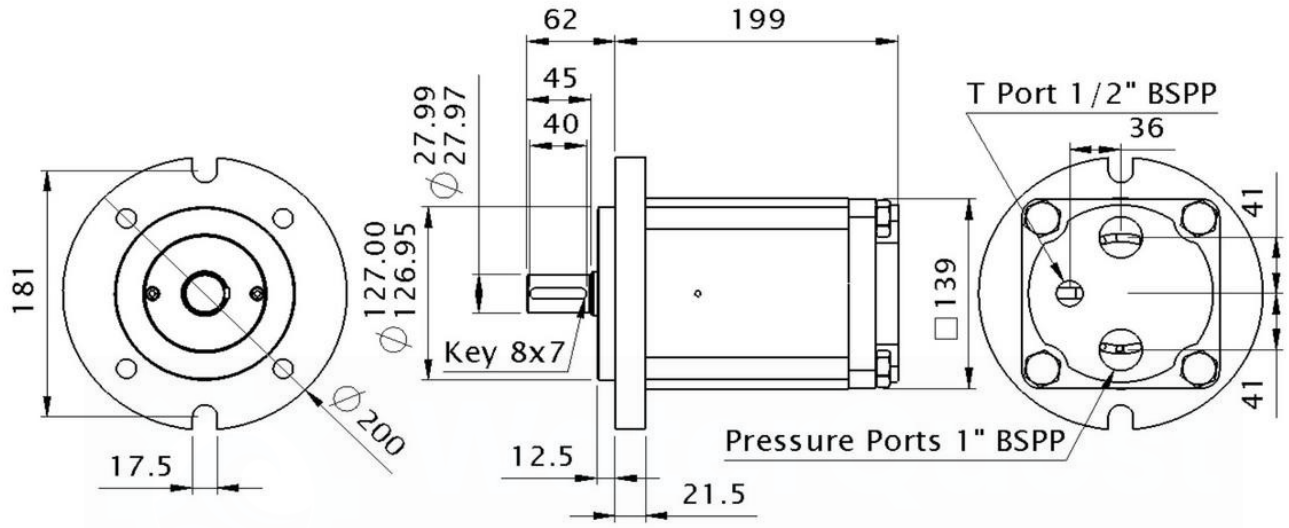


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

PB160-51W, PB160-63W and PB160-70W boosted P60 dimensional drawing, 1 inch BSPP pressure ports, 1/2 inch BSPP T port

PB160-51W, PB160-63W & PB160-70W



PC160, PA160 and PPC160 unboosted P60 dimensional drawing, 1 inch BSPP pressure, 1.5 inch BSPP T port, PPC adds M10 holes on 155mm PCD

PC160-51W, PC160-63W & PC160-70W

PA160-51W, PA160-63W & PA160-70W

PPC160-51W, PPC160-63W & PPC160-70W (O ring face seal and M10 holes only added on PP Option)

